

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 12	Team No-14	Project Title- Smart Attendance System Using Face Recognition
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1. Abstract

Attendance management in classrooms and offices is often performed manually, making the process time-consuming, repetitive, and prone to errors. The proposed **Smart Attendance System Using Face Recognition** aims to automate attendance by identifying registered students or employees through real-time facial recognition. The system is developed as an Android application using **Kotlin** and **CameraX** for capturing live camera input. **Google ML Kit Face Detection** is used to detect and locate faces, while **TensorFlow Lite** is used for face recognition and matching with enrolled facial data. During registration, the user's details and facial information are enrolled into the system for future identification. When an attendance session is active, the camera continuously detects faces and matches them against the registered profiles. Upon successful recognition, attendance is automatically recorded with the corresponding date and time. **Firebase Firestore** is used to store and synchronize attendance records and user information in the cloud. The system also provides an administrative interface for user management, attendance monitoring, reports, and analytics. The proposed solution reduces manual effort, minimizes attendance errors, and provides an efficient and scalable approach to automated attendance management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090151	N.Satwika	
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3. Frameworks / Platforms / Software Used

Android Studio – Development environment

Kotlin – Android application development

CameraX – Real-time camera capture and processing

Google ML Kit Face Detection – Face detection and facial feature analysis

TensorFlow Lite – Face recognition and face matching

Firebase Authentication – User/admin authentication

Firebase Firestore – Cloud database for user and attendance records

Firebase Storage – Storage of enrolled face images/data

Material Design – Android user interface development

Guide Name and Signature (remarks if any) _____